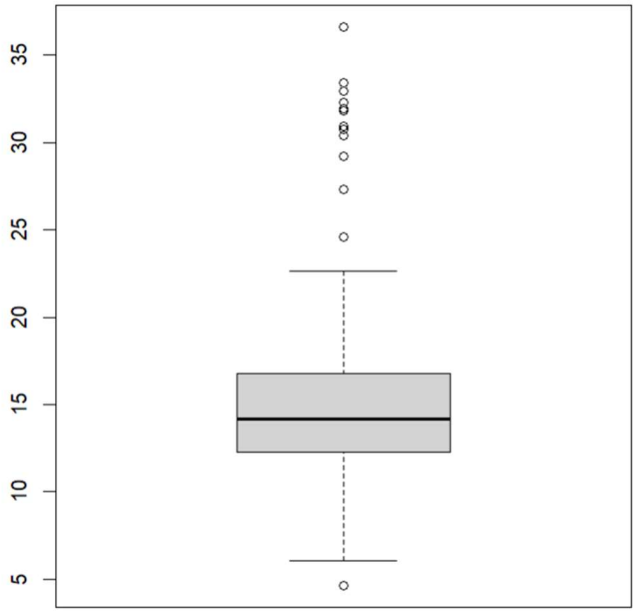
BIOSTAT 702: Exercise 3.1
One Sample Inference for a Continuous Outcome
Fall 2026

Question 1

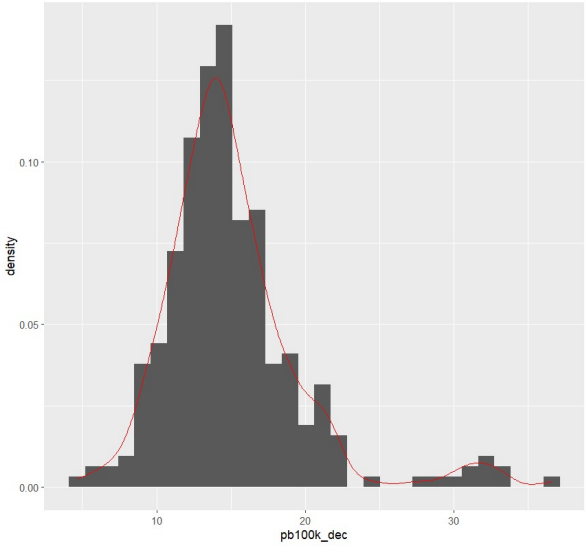
The dataset ultrarunning_altered.csv contains the same variables and research goals as the ultrarunning dataset we have encountered before, except now, some of the ultramarathon times have been altered slightly for learning purposes.

1. Create a box plot of the outcome variable, pb100k_dec. Describe the plot.



Most ultramarathon times fall between 12 and 17 hours, with a median of about 14 hours. There are several high outliers beyond 25 hours and one low outlier. The longer upper whisker and numerous high outliers suggest that the distribution is right-skewed.

2. Create a histogram, and overlay a kernel density plot. This is one way to generate a smoothed version of the histogram. Describe the plot.



The times are concentrated around 10-18 hours, with a peak near 14 hours. The histogram and smoothed density curve show a long right tail, with a few times above 30 hours. The distribution appears right- skewed.

3. Calculate the skewness of the ultramarathon times. Is this value consistent with the histogram? Based on the histogram and skewness, would you describe the distribution as approximately symmetric or skewed?

```
> library(moments)
> moments::skewness(ultra_altered$pb100k_dec)
[1] 1.66582
>
```

The skewness is 1.67, which is consistent with histogram's long right tail. The distribution is right-skewed rather than approximately symmetric.

Question 2

1. Let's explore the representativeness of the mean in this sample. Calculate the mean and standard deviation (i.e., a standard descriptive presentation when data are normally distributed). Calculate the median and interquartile range (i.e., a standard descriptive presentation when the data are not normally distributed). How different are the mean and median?

```
> sd(ultra_altered$pb100k_dec)
[1] 4.684084
>
```

```
> summary(ultra_altered$pb100k_dec)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   4.66   12.28   14.16   15.05   16.78   36.60
>
```

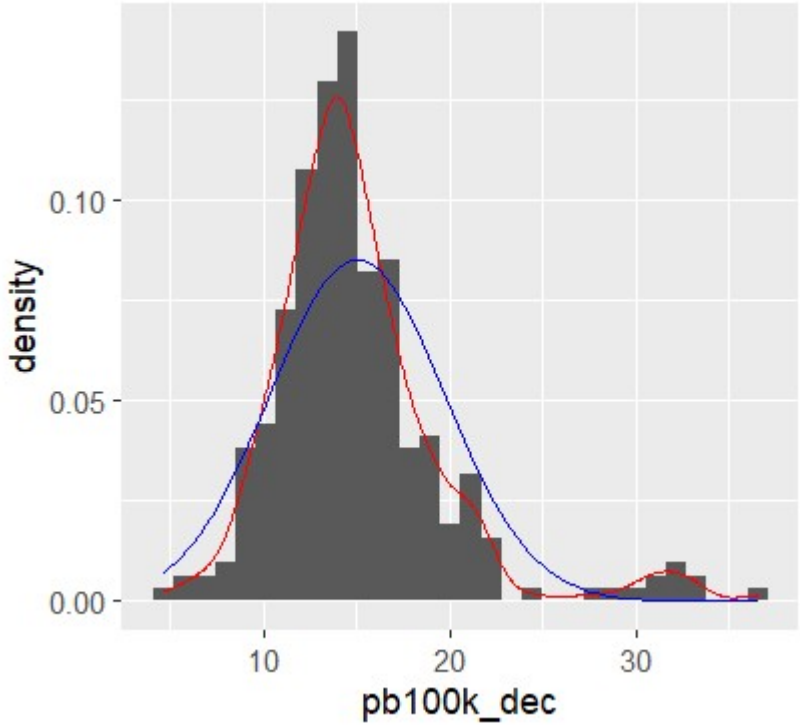
The mean ultramarathon time was 15.05 hours (SD 4.68), while the median was 14.16 hours (IQR 4.50). The mean was about 0.89 hours higher than the median, consistent with the longer times in the right tail of the distribution.

2. What percentage of observations are above the mean? If the distribution is symmetric, this should be 50%, except for the impact of sampling variability.

```
> mean(ultra_altered$pb100k_dec > mean(ultra_altered$pb100k_dec))
[1] 0.3993056
>
```

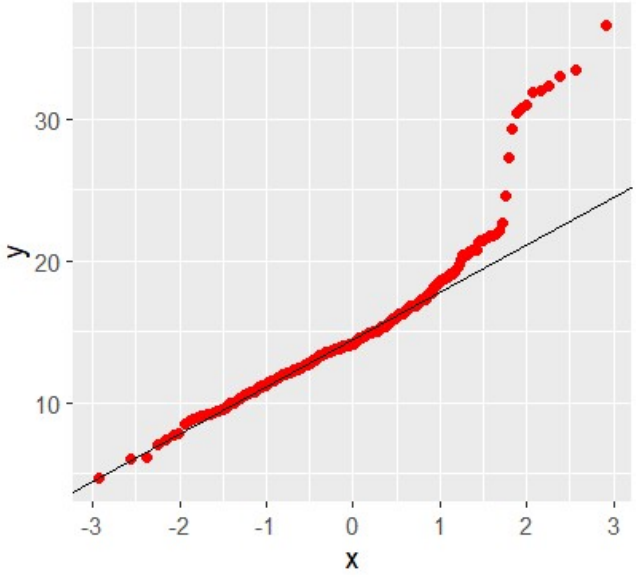
About 39.9% of the observations were above the mean. This is below 50%, which is consistent with the right-skewed distribution.

3. Append a normal density to your histogram, with the mean and standard deviation equivalent to those in the sample. Describe what you see.



Compared with the normal curve, the observed distribution has a sharper peak and a longer right tail. The normal curve does not capture the few very long times, so the data do not appear normally distributed.

4. Create a Q-Q plot comparing the distribution of ultramarathon times to normality. In plain English, explain what a Q-Q plot does and what you found.



A Q-Q plot compares the quantiles of observed data with those expected from a normal distribution. Most points in the middle follow the reference line, but the points at the upper end rise well above it. This suggest a heavier right tail than a normal distribution, so the times are not approximately normal.

Question 3

1. The investigator hypothesizes that the "average" ultramarathon time of their target population is 14 hours. It isn't clear whether they are referring to the median or the mean. Perform a t-test using this null hypothesis value. Is this testing the mean or the median? Write an interpretation of the result.

```
        One Sample t-test

data:  ultra_altered$pb100k_dec
t = 3.7862, df = 287, p-value = 0.0001863
alternative hypothesis: true mean is not equal to 14
95 percent confidence interval:
 14.50178 15.58831
sample estimates:
mean of x
 15.04504

>
```

The one sample t-test examines whether the population mean time is 14 hours. The sample mean was 15.05 hours, and the p-value was 0.00019. This provides strong evidence that the population mean differs from 14 hours. The sample estimate suggests it is higher.

2. Based on the sample size and distribution of the data, is the one-sample t-test performed above invalid? Explain why or why not.
No, the one sample t-test is not necessarily invalid. The times are right-skewed, the sample is large (n = 288). With a large sample, the sampling distribution of the mean is approximately normal, so the t-test can be used to test the mean. The unusually long times may affect the mean, so that limitation should be kept in mind when interpreting the result.

3. Would you choose a non-parametric test instead of the t-test? Why or why not?
I would keep the t-test because the question concerns the population mean, and the sample size is large (n = 288). Although, the times are right-skewed, a non-parametric test would address a different question about the center of the distribution. If the investigator meant the median by "average", I would consider a sign test instead.

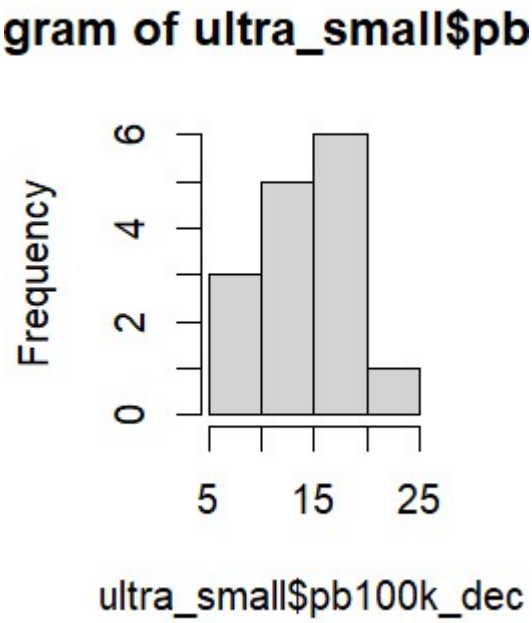
Question 4

1. Create a new dataset by taking a random sample of 15 participants from the altered ultrarunning dataset. Set the seed before you sample, for reproducibility purposes. Report the sample mean and sample median.

```
> mean(ultra_small$pb100k_dec)
[1] 14.46358
> median(ultra_small$pb100k_dec)
[1] 14.45775
>
```

In the random sample of 15 participants, the mean ultramarathon time was 14.46 hours and the median was 14.46 hours.

2. Create a histogram of the sampled ultramarathon times. Briefly describe the distribution. Does the normality assumption appear reasonable?



Most times in the sample fall between 10 to 20 hours, with the highest frequency around 15-20 hours. The histogram is not clearly symmetric or bell-shaped. With only 15 observations, it is difficult to assess normality confidently, so the normality assumption is questionable.

3. Perform a t-test using same null hypothesis as in Question 3. Report the p-value and state your conclusion.

```
> t.test(ultra_small$pb100k_dec, mu = 14)

        One Sample t-test

data:  ultra_small$pb100k_dec
t = 0.41825, df = 14, p-value = 0.6821
alternative hypothesis: true mean is not equal to 14
95 percent confidence interval:
 12.08635 16.84081
sample estimates:
mean of x
 14.46358
```

The one sample t-test gave a p-value of 0.6821. We do not reject the null hypothesis that the population mean is 14 hours because the sample does not provide sufficient evidence of a difference.

4. Perform a sign test using this null hypothesis value. Report the p-value and state your conclusion. Is this testing the mean or the median?

```
        Exact binomial test

data:  greater_median and trials
number of successes = 8, number of trials = 15, p-value = 1
alternative hypothesis: true probability of success is not equal to 0.5
95 percent confidence interval:
 0.2658613 0.7873333
sample estimates:
probability of success
             0.5333333
```

The sign test found that 8 of the 15times were above 14 hours (p = 1). We do not reject the hypothesis that population median is 14 hours, the sample provides no evidence that the median differs from 14 hours.

5. Perform a Wilcoxon signed rank test using this null hypothesis value. Report the p-value and state your conclusion. Is this testing the mean or the median?

```
        Wilcoxon signed rank exact test

data:  ultra_small$pb100k_dec
V = 69, p-value = 0.6387
alternative hypothesis: true location is not equal to 14
95 percent confidence interval:
 11.92583 17.00529
sample estimates:
(pseudo)median
      14.43846
```

The Wilcoxon signed-rank test gave a p-value of 0.6387. We do not reject the null hypothesis that the distribution is centered at 14 hours. This test can be interpreted as a test of the median if the distribution of differences from 14 hours is approximately symmetric.

6. Compare the results of the three tests. Did they all reach the same conclusion? If not, why might they differ? Which test would you feel most comfortable using for your sample, and why?

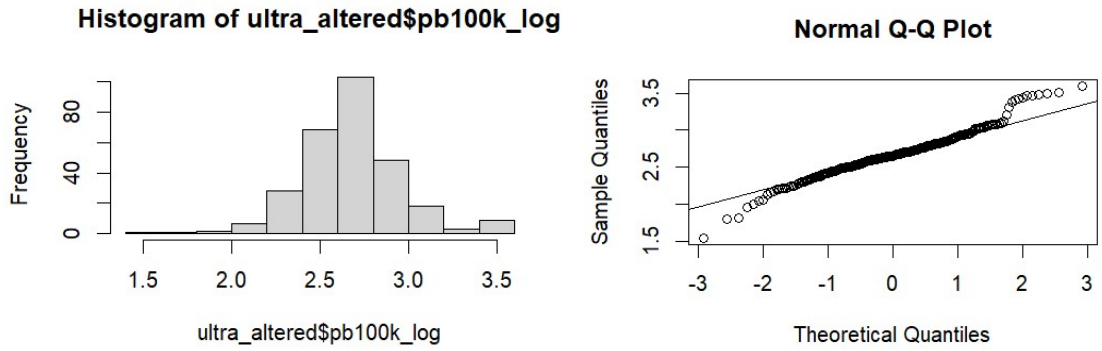
All three tests reached the same conclusion: none provided sufficient evidence of a difference from 14 hours (t-test p = 0.6821, sign test p = 1, and signed-rank test p = 0.6387). Their p-values differ because they use different aspects of the data and do not always test the same measure of center. For this small sample, I would be most comfortable with the sign test if the question concerns the median, because the histogram does not clearly support normality or symmetry.

Question 5

Thus far, we have used a one-sample t-test to make inference about the arithmetic mean ultramarathon time. For right-skewed outcomes, another quantity that is sometimes of interest

is the geometric mean, which can be obtained by taking the mean on the log scale and then exponentiating.

1. Take the log of ultramarathon times (i.e., to the base e). Examine the distribution of the transformed variables using graphical techniques. Describe what you see.



After the log transformation, the distribution appears more symmetric than the original times. In the Q–Q plot, the middle values lie fairly close to the reference line, but both tails deviate from it, especially the upper tail. The log-transformed times are closer to normality, though they are not perfectly normal.

2. Create a 95% confidence interval for the mean of the log-ultramarathon times. Exponentiate the point estimate and the upper and lower endpoints and interpret.

```
> exp(est_log)
[1] 14.4213
> exp(ci_log_lower)
[1] 13.94777
> exp(ci_log_upper)
[1] 14.9109
>
```

The estimated geometric mean ultramarathon time was 14.42 hours. The 95% confidence interval for the population geometric mean was 13.95 to 14.91 hours. I calculated the confidence interval on the log scale and exponentiated both endpoints to express it in hours.

3. Compare the estimate obtained here to the sample mean from Question 3. Why are they different

```
        One Sample t-test

data:  ultra_altered$pb100k_log
t = 157.33, df = 287, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 2.635320 2.702092
sample estimates:
mean of x
 2.668706
```

The geometric mean of 14.42 hours was lower than the arithmetic mean of 15.05 hours. The few long times in the right tail increase the arithmetic mean, while they have less influence when the times are averaged on the log scale.